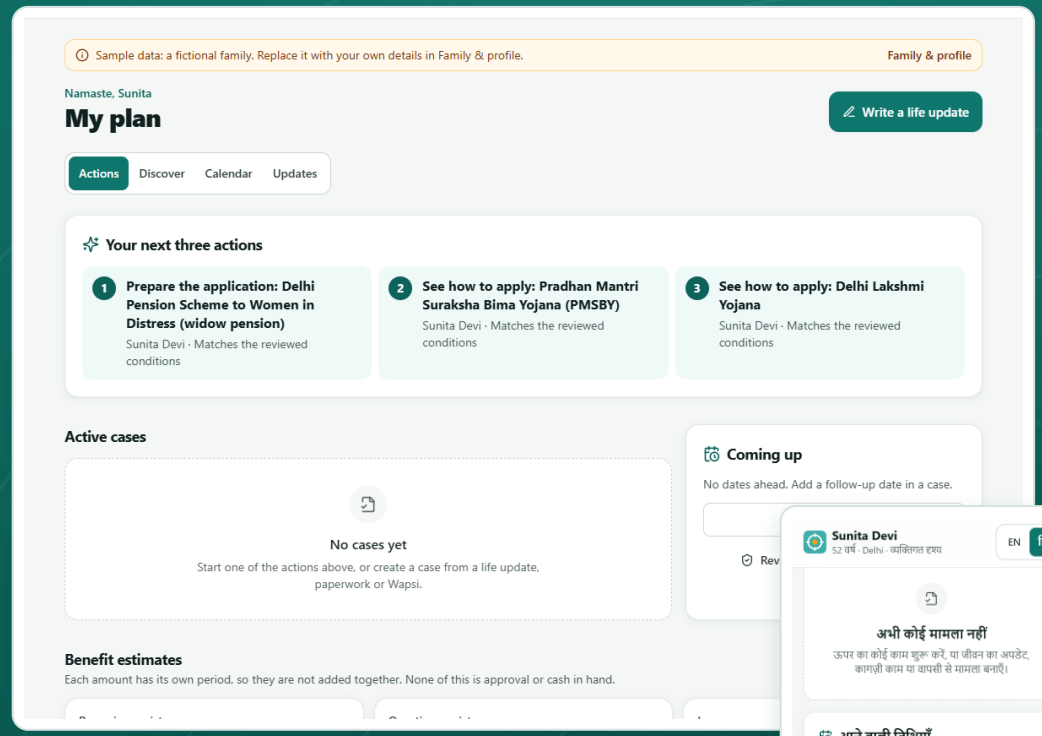
HackIIITD '26 · Esya · Final submission

JAAGRUK

जागरूक Janta's AI Agent for Government Rights, Updates & Knowledge

Every rupee your family is owed — found, proven, claimed.

A Hindi/English app that checks each entitlement against official rules with a Z3 solver, turns life changes into ready-to-act cases, and prepares unclaimed-money recovery on the live government web through TinyFish.





Z3-checked eligibility · 6 TinyFish products · Live .gov.in portals · हिन्दी + English

THE TEAM

Ayush Awasthi (AA) — AI/ML & agents

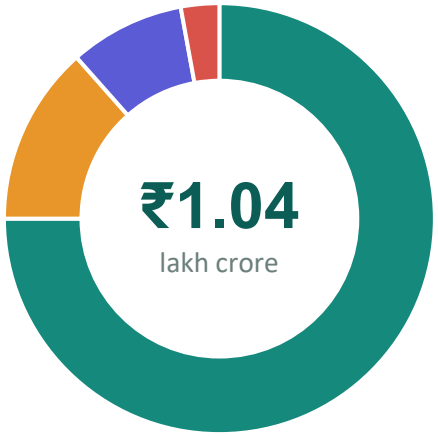
Saif Nasim (SN) — Backend & TinyFish

Sagar Hundia (SH) — Frontend & UI/UX

₹1.04 lakh cr unclaimed in India

Families miss money and benefits they are already owed

UNCLAIMED IN INDIA



- Bank deposits **₹78k cr**
- Insurance **₹14k cr**
- Dividends **₹9k cr**
- Mutual funds **₹3k cr**

Source: PIB, Dec 2025

5,000+

government schemes — citizens must search many department websites and guidelines to check eligibility

MeitY · myScheme

74%

of PM-JAY households surveyed did not know how to claim a benefit they already had

Haqdarshak survey, 6 states

SUNITA, 52, DELHI · HUSBAND DIED LAST MONTH

Rules are scattered

Each scheme has its own age, income and residence rules, in legal English across portals and PDFs.

Money is hidden

Each unclaimed-money portal covers one asset type, with its own form, login, OTP or CAPTCHA.

Finder ≠ payer

UDGAM, MITRA and IEPF find a record — the bank, AMC or company actually pays.

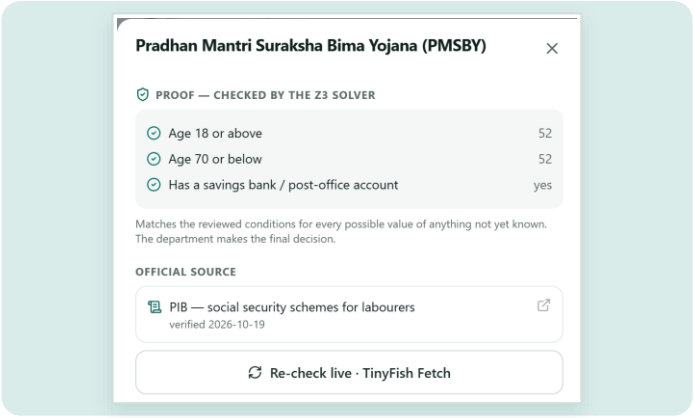
Deadlines move

Her daughter's scholarship renewal changes on boards nobody watches.

People claim when someone helps: ₹5,777 cr went back to 22.95 lakh claimants in three months of help camps (PIB, Mar 2026). Portals list information — nobody watches, matches, verifies and acts for one family.

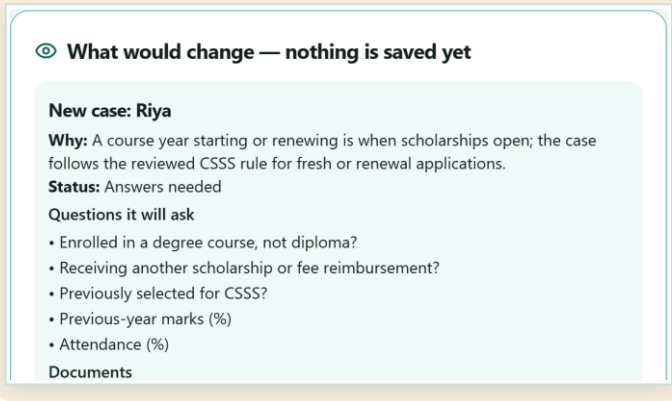
Find, check, prepare — and the family approves every step

One phone-first app for an Indian family's schemes, scholarships, exams, insurance and its own unclaimed money.



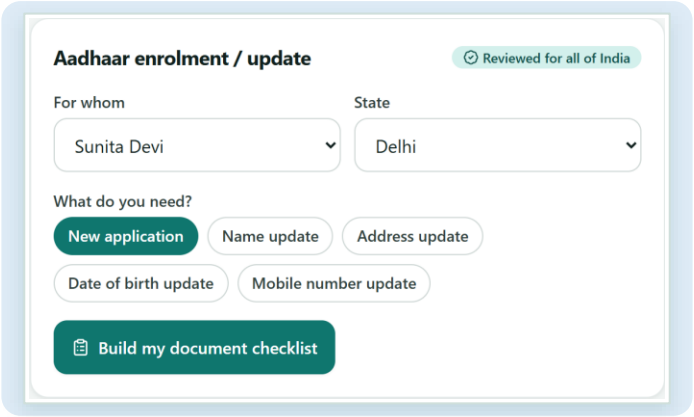
My plan

- Next 3 actions + benefit estimates by type
- "Why?": Z3 checks every clause, with source and date
- "Re-check live" reads today's page (TinyFish Fetch)



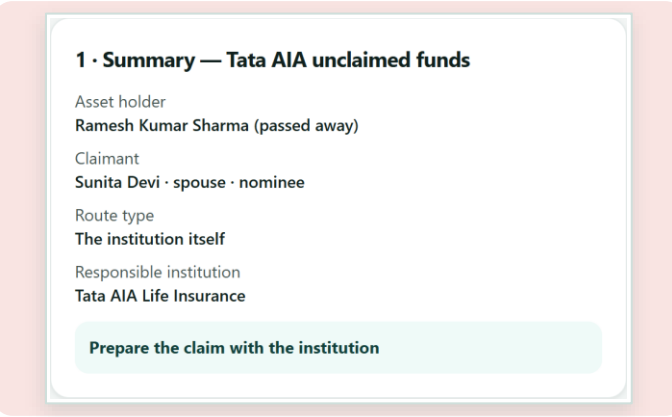
Life update

- Write or say: "Riya got into second year"
- Preview the new case, questions, documents — nothing saved yet
- Apply refused if plan or rules changed



Paperwork

- Minimum checklist for this person, service and state
- From reviewed official routes — no invented lists
- Anything else: TinyFish Search finds .gov.in guidance



Wapsi

- Case first: holder, claimant, relation, claim capacity
- Automated check (LIC, Tata AIA) or official site for OTP/CAPTCHA
- Claim prepared with the institution that pays

Reasons, not guesses · **Nothing submitted for you** · **Never sees an OTP or password** · **हिन्दी + English, voice**

A light, testable stack built for the live government web

Frontend	**React 18** component UI	**TypeScript** typed end to end	**Vite** fast build, lazy screens	**Tailwind CSS** 48 px touch targets	**Installable PWA** हिन्दी / English, voice
Backend	**Python 3.12** typed services	**FastAPI** REST + SSE job streams	**Job ledger** restart-safe jobs	**SQLite (WAL)** call log, helper links	**FTS5** 1,135-scheme search
Reasoning & ML	**Z3 SMT solver** three-valued proofs	**z3.Optimize** fewest docs to unlock	**Name matcher** fuzzy + phonetic	**Exam forecaster** circular statistics	**Claude** optional: cited answers
Web layer	**TinyFish Agent** run-sse, schemas	**TinyFish Browser** CDP + Playwright	**Fetch + Search** free public reads	**Monitor** + Research, cited	**India exit proxy** for blocked portals

QUALITY — pytest · 103 offline tests · Frontend tests · Live adapter qualification · Recorded replay offline · Error boundary + self-heal

Why this stack: rules are data, a solver proves each answer, every web call is logged — and the whole app runs offline on recorded real runs.

One rules registry, one job ledger, one TinyFish harness

CLIENT

Citizen or CSC

Phone or desktop, Hindi or English, voice input

React PWA

Workspace, cases and journal stay in this browser; lazy screens, error boundary

API

FastAPI

My plan · Life update · Paperwork · Wapsi · Ask · Under the hood; private responses no-store

Job ledger + SSE

Idempotent starts, cancel, live progress; “uncertain” after a restart

SERVICES

Rules registry → Z3

One rules_version for plan, cases, Discover, Ask

Recovery cases

Bound to case, holder, route; strict outcomes

Updates + Discover

Notice boards, deadlines, 1,135 scheme pages

WEB LAYER

TinyFish harness

Plan → run → verify → self-heal → hand-off; every call logged, public reads replayable

Official web

.gov.in · myScheme · NSP · LIC · Tata AIA · notice boards (India exit IP)

SQLite (WAL) TinyFish call log · job ledger · 24-hour helper links behind a hashed token · reviewed rule changes · no IDs, passwords or OTPs stored

All six TinyFish products, behind a harness that never guesses

Agent · run-sse
Tata AIA check · second opinion
5 steps · 66 s · ≈ $0.08

Browser · CDP
LIC playbook · mySchemes pages
LIC in 38.6 s · ≈ $0.002

Fetch
Updates scan · re-check · deadlines
all boards in one call · free



Search
Discover · paperwork · new schemes
≈ 1 s · free

Monitor
Always-on board watching
private webhook · operator-enabled

Research
Ask → live research, cited
cited .gov.in answers

THE HARNESS — EVERY CALL

1. **Plan** pick the qualified playbook or agent
2. **Run** India exit IP, hard caps 180 s / 30 steps
3. **Verify** read the portal's own result area
4. **Self-heal** one bounded fallback, only before submit
5. **Hand-off** OTP / CAPTCHA → citizen on the official site

STRICT OUTCOMES — EVERY CHECK ENDS IN ONE OF THESE

- Possible record
- No match
- Citizen step
- Missing input
- Unavailable
- Failed
- Uncertain
- Cancelled

CAPTCHA, outages and timeouts never become “no match”. A record is never treated as ownership, and amounts are never summed.

What the live government web taught us — and how we prove it

Portal	What we found live	What we changed
LIC	The default network returned proxy_error	Every TinyFish browser and agent exits from India
SBI Life	Agent said “COMPLETED, not found” — it had hit a CAPTCHA	Blocked ≠ no match; the citizen completes it on the official site
IEPF / MCA	Search moved behind a login; an uncapped agent ran 83 steps	Hard caps on every run; official-site route
Delhi widow pension	Official text: income “less than ₹1,00,000”	Extractor fixed; Z3 proves our rule matches the live page
Agent outputs	found: true alongside “Found 0 unclaimed amounts”	Strict portal-specific reader + a test from the recorded case

Research sources: PIB (Dec 2025, Mar 2026) · MeitY mySchemе · Haqdarshak PM-JAY survey · RBI UDGAM · official portal pages read with TinyFish

Live tests use only a fictional identity (“Jaagrukdemo Testcase”). They show how portals behaved on the day — not a paid claim.

103

backend tests pass offline — recovery, jobs, rules registry, eligibility, life updates

2 / 2

automated Wapsi adapters live-qualified on 25 Oct 2026 — LIC 38.6 s, Tata AIA 66 s

0.979

name-matcher F1 on synthetic Indic name pairs; exact matching finds only 14% of true variants

Cheap to run, clear to earn from, ready to scale

WHAT A CHECK COSTS

≈ $0.002

LIC playbook check on a TinyFish Browser

≈ $0.08

Tata AIA check with the TinyFish Agent

₹0

Search, Fetch and Research reads

BUSINESS MODEL

CSC / NGO console

₹199 per centre a month

CSCs already earn ₹30 per scholarship application; 1% of 5.16 lakh CSCs ≈ ₹1.23 cr a year

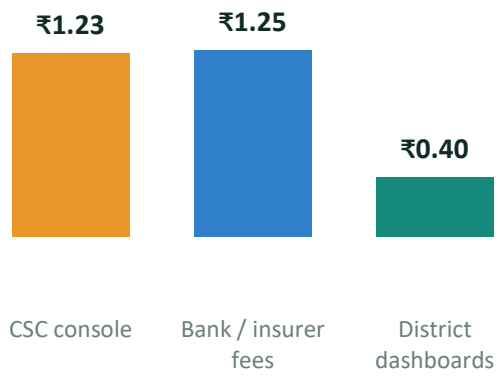
Banks & insurers

1% of a settled claim, capped at ₹500

Paid only after a claim settles; the citizen pays nothing for recovery

Year-2 projection ≈ ₹2.9 cr

conservative · ₹ crore a year



WHAT'S NEXT

- Every state's schemes through the same reviewed-rules pipeline
- Bhashini voice and more Indian languages
- More banks and insurers as live-qualified adapters (IRDAI Bima Bharosa list)
- CSC operator mode: one operator serves a whole village
- SQLite → Postgres, jobs → a queue: drop-in upgrades

Thank you

JAAGRUK जागरूक

Every rupee your family is owed — found, proven, claimed.

Ayush Awasthi · Saif Nasim · Sagar Hundia

4 destinations: My plan, Life update, Paperwork, Wapsi

6 TinyFish products on live .gov.in portals

103 automated tests, plus live adapter qualification

Run it: `python run.py` · offline: `python run.py --replay` · evidence: `docs/VALIDATION.md`